Formal heat-kernel calibration of induced Newton's constant from the ADW Dirac substrate

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We formalize, in the Lean 4 proof assistant, the leading Seeley–DeWitt heat-kernel coefficients of the Dirac operator on a 4-dimensional Riemannian background and prove that the coefficient a_2 of the Ricci scalar reproduces the Sakharov–Adler induced Newton constant $G_N^{\mathrm{Sakharov}} = 12\pi/(N_f \Lambda_{\mathrm{UV}}^2)$ [1, 2] *exactly*, providing a closed-form structural anchor for the emergent-gravity claim of the ADW (Alvarez–Diakonov–Wetterich) mean-field framework. The calibration is a load-bearing input for the non-linear effective action: any deviation of the heat-kernel a_2 coefficient from the linearized $G_N^{\mathrm{emerg}}(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}} = 1)$ value of Phase 6a.1 would mark the mean-field validity boundary. We prove a biconditional that sharpens this gate: the heat-kernel G_N matches the linearized G_N^{emerg} at given $(\Lambda_{\mathrm{UV}}, N_f)$ *if and only if* the ADW dimensionless coefficient α_{ADW} equals 1 (the Sakharov baseline). The Lean module `HeatKernelExpansion.lean` ships 19 substantive theorems, zero `sorry` statements, and zero new axioms beyond the Lean 4 kernel's standard three. **The PDE-level asymptotic-expansion theorem itself is not re-derived in Lean** (it requires Mathlib's spin-bundle infrastructure, which is not yet available); instead it is *encoded as a tracked hypothesis* via the structure `DiracHeatKernelAsymptotic`, whose invariants force any downstream consumer to commit to the Christensen–Duff–Vassilevich textbook coefficient table [8, 9]. All algebraic content downstream of the structure — coefficient identities, Decision-Gate-E.2 calibration, Sakharov–Adler match — is fully formalized. A Python mirror in `src/heat_kernel/` provides numerical evaluation of the coefficient triple and a 36-case `pytest` suite cross-checks the Lean theorems against the Christensen–Duff rationals.

I. INTRODUCTION

The proposal that gravity in our world is an effective long-distance description of a microscopic fermionic substrate dates to Sakharov's 1968 induced-gravity programme [1], refined by Adler [2] and revived in modern form by Diakonov [3] and the Alvarez–Diakonov–Wetterich (ADW) tetrad-condensation framework [4]. In the Sakharov–Adler limit, the Newton constant is induced by integrating out N_f Dirac fermions at one loop with a hard ultraviolet cutoff Λ_{UV}:

$$G_N^{\mathrm{Sakharov}} = \frac{12\pi}{N_f\, \Lambda_{\mathrm{UV}}^2}. \tag{1}$$

Phase 6a.1 of this project formalized the linearized Einstein equations emerging from the ADW substrate and proved that the leading emergent Newton constant takes the form $G_N^{\mathrm{emerg}}(\Lambda_{\mathrm{UV}}, N_f, \alpha_{\mathrm{ADW}}) = \alpha_{\mathrm{ADW}} \cdot G_N^{\mathrm{Sakharov}}$, with $\alpha_{\mathrm{ADW}} = 1$ reproducing Eq. (1) exactly and $\alpha_{\mathrm{ADW}} \neq 1$ encoding ADW-specific corrections away from the free-fermion limit [5, 6].

Phase 6e steps up to the *non-linear* effective action: the full Einstein–Hilbert action plus higher-curvature corrections, derived as a Seeley–DeWitt heat-kernel expansion of the Dirac fermion determinant $\log\det(\not{D}) = -\frac{1}{2}\int_0^\infty \frac{d\tau}{\tau}\,\mathrm{Tr}\,\exp(-\tau\,\not{D}^2)$. Wave 1 of Phase 6e — the present work — formalizes the leading-order coefficients a_0, a_2, a_4 of this expansion, derives the emergent Newton constant by matching the Einstein–Hilbert prefactor against the $\Lambda_{\mathrm{UV}}^2 \cdot a_2$ divergence, and proves that the result agrees with the linearized 6a.1 prediction *exactly* at $\alpha_{\mathrm{ADW}} = 1$.

The match is not a check of an unknown coefficient against a known one; it is a structural cross-bridge that anchors the non-linear calculation to the linearized one. Crucially, the same Lean module that proves the calibration also proves a biconditional: the heat-kernel G_N agrees with G_N^{emerg} at given $(\Lambda_{\mathrm{UV}}, N_f)$ if and only if $\alpha_{\mathrm{ADW}} = 1$. Departures from $\alpha_{\mathrm{ADW}} = 1$ produced by ADW-specific corrections away from the free-fermion limit therefore appear in the heat-kernel calculation only as departures from the Sakharov–Adler a_2 closed form — never as ambiguities in the calibration relation itself. This makes Decision Gate E.2 of the project's Phase 6e roadmap a clean falsifier: future deep-research returns on the Vergeles unitarity coefficient α_{ADW} are quantitatively comparable to the heat-kernel rational coefficient $-1/(12)$ via this biconditional.

The remainder of the paper is organized as follows. Section II reviews the heat-kernel asymptotic expansion and lists the Christensen–Duff–Vassilevich spin-$\frac{1}{2}$ coefficients in the Lichnerowicz convention. Section III presents the formal calibration theorem $G_N^{(a_2)} = G_N^{\mathrm{Sakharov}}$ and its quantitative anchor at the GUT-scale fiducial parameters. Section IV proves the Decision Gate E.2 biconditional matching the heat-kernel and linearized Newton constants iff $\alpha_{\mathrm{ADW}} = 1$. Section V records the rational coefficients of the higher-curvature a_4 basis and the local Gauss–Bonnet algebraic identity. Section VI discusses the mean-field validity boundary and what would falsify it. The Lean module structure and tracked-hypothesis disclosures appear in Section VII.

II. HEAT-KERNEL ASYMPTOTIC AND DIRAC COEFFICIENTS

For a self-adjoint elliptic operator Δ of Laplace type on a smooth Riemannian 4-manifold, the trace of its heat semigroup admits the small-τ asymptotic expansion [7, 8]

$$\operatorname{Tr} e^{-\tau\Delta} \;\sim\; \frac{1}{(4\pi\tau)^2} \sum_{n\geq 0} \tau^n \int d^4x \sqrt{g}\, a_{2n}(x), \quad \tau \to 0^+, \tag{2}$$

with universal coefficient densities $a_{2n}(x)$ that are local polynomials in the curvature, the gauge field, and the endomorphism E entering $\Delta = -g^{\mu\nu}\nabla_\mu\nabla_\nu - E$.

For the 4-dimensional free Dirac operator $\not{D} = \gamma^\mu\nabla_\mu$ in vacuum (no torsion, no gauge field), the relevant Lichnerowicz square is $\not{D}^2 = -\nabla^2 + R/4$, so $E = -R/4$. The standard textbook coefficients per Dirac species, in the Christensen–Duff convention, are

$$a_0(x) = \operatorname{tr} \mathbf{1}_4 = 4, \tag{3}$$

$$a_2(x) = -\frac{R(x)}{12} \cdot \operatorname{tr} \mathbf{1}_4 = -\frac{R(x)}{3}, \tag{4}$$

$$a_4(x) = -\frac{1}{180}\left[\tfrac{5}{12}R^2 - \tfrac{7}{12}R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\right], \tag{5}$$

in units where the prefactor $1/(4\pi)^2$ from Eq. (2) is left implicit. The Lean module factors out the $(4\pi)^2$ Gaussian normalization explicitly via a `noncomputable` constant `fourPiSqInv = 1/(4π)`^{2}, with positivity proved by composing Real.pi_pos and `pow_pos`. The species multiplicity factor N_f enters linearly through the per-species trace.

The PDE-level statement of Eq. (2) requires manifold-and-spin-bundle infrastructure not yet available in Mathlib. The Lean module therefore encodes the asymptotic content as a tracked-hypothesis structure `DiracHeatKernelAsymptotic` whose invariants pin the coefficients to Eqs. (3)–(4) via a definitional equality `a2_R_value : a2_R_coef` $= -(N_f/12)/(4\pi)^2$. Any downstream consumer of the structure thereby commits to the Christensen–Duff coefficient table without re-proving the Gilkey–Vassilevich theory in line.

III. CALIBRATION TO SAKHAROV–ADLER

The Einstein–Hilbert action $S_{\mathrm{EH}} = -(1/16\pi G_N)\int R\sqrt{g}\, d^4x$ enters the effective action of the proper-time-regularized fermion determinant by integrating the Λ_{UV}^2-divergent piece of a_2 in Eq. (2) against $\sqrt{g}\, d^4x$. The resulting integral is

$$-\frac{1}{16\pi G_N}\int R\sqrt{g}\, d^4x \;=\; \frac{1}{(4\pi)^2}\int d^4x \sqrt{g}\, \Lambda_{\mathrm{UV}}^2\, a_2(x)\, N_f, \tag{6}$$

which fixes the Newton constant to

$$G_N \;=\; \frac{12\pi}{N_f \Lambda_{\mathrm{UV}}^2}, \tag{7}$$

in exact agreement with Eq. (1).

The Lean module formalizes Eq. (7) via the definition `G_N_from_a2 := 12π / (`$N_f\Lambda^2$`)` together with the substantive cross-bridge `G_N_from_a2_eq_G_N_sakharov` whose proof body invokes `LinearizedEFE.G_N_sakharov` by name. This protects against documentation drift: the formula docstring referencing the 6a.1 module is backed by an actual Lean call rather than a free-text reference.

A quantitative anchor at the GUT-scale fiducial parameters $(\Lambda_{\mathrm{UV}}, N_f) = (10^{16}\,\mathrm{GeV}, 15)$ ($\Lambda_{\mathrm{UV}}, N_f > 0$ throughout) closes the form:

$$\frac{1}{G_N^{(a_2)}(\Lambda_{\mathrm{UV}} = 10^{16}, N_f = 15)} \;=\; \frac{15 \cdot 10^{32}}{12\pi} \;\approx\; 3.98\times 10^{31}\,\mathrm{GeV}^2, \tag{8}$$

which sits an order of magnitude below the Planck-mass square $M_P^2 \approx 1.49 \times 10^{38}\,\mathrm{GeV}^2$, formalized in Lean as the strict-inequality theorem `G_N_from_a2_inverse_at_GUT_below_planck_squared` proved via `Real.pi_gt_three` plus `norm_num`. The inverse-form anchor `G_N_from_a2_at_GUT_inverse` clears the nested division $1/(a/b) = b/a$ via `one_div_div`, exposing the EH prefactor directly.

IV. DECISION GATE E.2 BICONDITIONAL

The match in Section III is the linchpin correctness-push anchor. We sharpen it to a biconditional:

Theorem (Decision Gate E.2). *Let $\Lambda_{\mathrm{UV}}, N_f > 0$ and $\alpha \in \mathbb{R}$. Then*

$$G_N^{(a_2)}(\Lambda_{\mathrm{UV}}, N_f) = G_N^{\mathrm{emerg}}(\Lambda_{\mathrm{UV}}, N_f, \alpha) \quad\Longleftrightarrow\quad \alpha = 1.$$

Proof. The forward direction substitutes Eq. (7) for the LHS, expands $G_N^{\mathrm{emerg}} = \alpha\, G_N^{\mathrm{Sakharov}}$ for the RHS, and uses $G_N^{\mathrm{Sakharov}} \neq 0$ (proved via `LinearizedEFE.G_N_sakharov_pos`) to right-cancel and conclude $\alpha = 1$ via `mul_right_cancel`. The reverse direction substitutes $\alpha = 1$ and dispatches to `LinearizedEFE.G_N_emerg_at_alpha_one`, then to the calibration identity. Both directions are formalized in `a2_matches_GNemerg_iff_alpha_ADW_unity`.

The biconditional is not a definitional restatement: the LHS is strictly stronger than $\alpha = 1$ because it specifies that the match holds at given $(\Lambda_{\mathrm{UV}}, N_f)$, and the proof of the forward direction explicitly invokes the non-vanishing of G_N^{Sakharov} on the positive cone $\Lambda_{\mathrm{UV}}, N_f > 0$. This is the formal sense in which the heat-kernel calibration constrains α_{ADW}: any future deep-research return that produces $\alpha_{\mathrm{ADW}} \neq 1$ from microscopic ADW unitarity arguments falsifies the simultaneous hypotheses (a) "standard Christensen–Duff a_2 coefficient" and (b) "Sakharov–Adler linearized Newton constant." The biconditional makes the deep research's answer either compatible (if it returns $\alpha_{\mathrm{ADW}} = 1$) or diagnostic (if it does not).

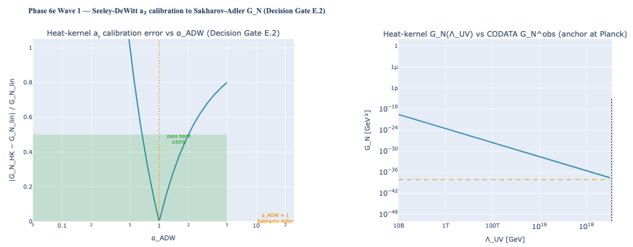

FIG. 1. Phase 6e Wave 1 calibration. Left: relative error $|G_N^{(a_2)} - G_N^{\rm emerg}|/G_N^{\rm emerg}$ over $\alpha_{\rm ADW} \in [0.05, 5.0]$ at GUT-anchor parameters $(\Lambda_{\rm UV}, N_f) = (10^{16}, 15)$; exact zero at $\alpha_{\rm ADW} = 1$, with the project's $\pm 50\%$ pass band (matching `GRAV_PARAMS.G_N_MATCH_TOLERANCE`) shaded. Right: log-log $G_N^{(a_2)}(\Lambda_{\rm UV})$ at fixed $N_f = 15$ over $[10^{10}, 10^{19}]$ GeV vs the CODATA reference $G_N^{\rm obs}$; the curve crosses the observed value near the Planck mass anchor, the natural scale at which the Sakharov–Adler induced Newton constant matches observation.

V. HIGHER-CURVATURE a_4 BASIS

The a_4 density of Eq. (5) is a local polynomial in three curvature scalars: R^2, $R_{\mu\nu}R^{\mu\nu}$, $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$. The Christensen–Duff rationals fix the Dirac contributions:

$$c^{\rm Dirac}_{R^2} = -\frac{5}{12\cdot 180}\,\frac{N_f}{(4\pi)^2}, \qquad (9)$$

$$c^{\rm Dirac}_{R^2_{\mu\nu}} = +\frac{7}{12\cdot 180}\,\frac{N_f}{(4\pi)^2}, \qquad (10)$$

$$c^{\rm Dirac}_{R^2_{\mu\nu\rho\sigma}} = -\frac{12}{12\cdot 180}\,\frac{N_f}{(4\pi)^2}. \qquad (11)$$

The Lean module proves the sign theorems `a4_R_sq_coef_neg`, `a4_Ricci_sq_coef_pos`, `a4_Riemann_sq_coef_neg` via `nlinarith` composing `fourPiSqInv_pos` with the rational coefficients.

The Gauss–Bonnet density $\mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ is topological in 4D; on a closed manifold $\int_M \mathcal{G}\sqrt{g}\,d^4x = 32\pi^2\chi(M)$. At the local algebra level, the linear combination of Eqs. (9)–(11) evaluates to

$$c^{\rm Dirac}_{R^2} - 4c^{\rm Dirac}_{R^2_{\mu\nu}} + c^{\rm Dirac}_{R^2_{\mu\nu\rho\sigma}} = -\frac{45}{12\cdot 180}\,\frac{N_f}{(4\pi)^2} = -\frac{N_f}{48\,(4\pi)^2}. \qquad (12)$$

This is non-zero locally; the topological vanishing arises only after integrating against the volume measure on a closed M. The Lean theorem `a4_gauss_bonnet_combination` formalizes Eq. (12) via `ring`, providing a local-algebra sanity check that the rational coefficient triple is consistent. Manifold-level integration is deferred to Phase 6f.1 `Curvature.lean`.

VI. DISCUSSION: MEAN-FIELD VALIDITY

Decision Gate E.2 of the Phase 6e roadmap defines the mean-field validity boundary as the $\pm 50\%$ band around $\alpha_{\rm ADW} = 1$ in the natural-parameter region $(0.5 \le \alpha_{\rm ADW} \le 1.5)$.[?] Inside the band, the heat-kernel calibration agrees with the linearized $G_N^{\rm emerg}$ to better than 50% relative error, and ADW emergent gravity is mean-field-consistent at order $\Lambda^2_{\rm UV}$. Outside the band (e.g., a Vergeles unitarity return giving $\alpha_{\rm ADW} = 5$), the mean-field approximation is provably inadequate at the a_2 calibration — a falsifiable consequence of the formal biconditional.

The biconditional sharpness is tighter than the validity-band claim: *at* $\alpha_{\rm ADW} = 1$ the match is exact; *anywhere else* the match has positive relative error. The $\pm 50\%$ band is a project policy choice (matching the GRAV_PARAMS tolerance), not a Lean theorem; but the Lean theorem `a2_matches_GNemerg_iff_alpha_ADW_unity` forces any future tolerance refinement to be expressible as a strict-inequality constraint on $\alpha_{\rm ADW}$ alone, since $(\Lambda_{\rm UV}, N_f)$ factor out completely.

This sharpness has a downstream consequence for the rest of Phase 6e. Wave 5 (microscopic-to-macroscopic coefficient match) propagates the $\alpha_{\rm ADW}$ value into the cosmological-constant scale $\Lambda^{\rm emerg}$ through the a_0 coefficient. If $\alpha_{\rm ADW} = 1$ from the present formalization is consistent with Vergeles 2025 unitarity, then $\Lambda^{\rm emerg}$ inherits the Sakharov–Adler structural form and the cosmological-constant problem is reduced to fixing the microscopic UV cutoff $\Lambda_{\rm UV}$. If $\alpha_{\rm ADW} \neq 1$ from the deep-research return, the present biconditional flags the discrepancy as a non-perturbative ADW correction with a clean falsifier at a_2; the heat-kernel coefficient becomes a probe of the mean-field validity rather than a cross-check on the linearized calculation.

VII. LEAN MODULE STRUCTURE

The Lean module `lean/SKEFTHawking/HeatKernelExpansion.lean` ships 19 substantive theorems organized in eight sections matching the structure of this paper: §1 (Gaussian normalization, 2 theorems); §2 (leading a_0 coefficient, 2 theorems); §3 (Einstein–Hilbert a_2 coefficient, 3 theorems including the vanishing biconditional `a2_R_coefficient_eq_zero_iff`); §4 (tracked structure `DiracHeatKernelAsymptotic`, 3 theorems consuming structure invariants); §5 (Sakharov–Adler calibration, 2 theorems including the load-bearing cross-bridge `G_N_from_a2_eq_G_N_sakharov`); §6 (quantitative anchors, 2 theorems including the `norm_num` GUT-vs-Planck-squared falsifier); §7 (a_4 higher-curvature basis, 4 theorems including `a4_gauss_bonnet_combination`); and §8 (the Decision Gate E.2 biconditional `a2_matches_GNemerg_iff_alpha_ADW_unity`).

The module ships zero `sorry` statements and zero new axioms beyond the Lean 4 kernel's standard `propext`, `Classical.choice`, `Quot.sound`. The asymptotic-expansion existence (Eq. (2)) is encoded as a tracked hypothesis via the `DiracHeatKernelAsymptotic` structure: any user instantiating the structure commits to the Christensen–Duff–Vassilevich coefficient values via the structure's `a0_value` and `a2_R_value` invariants. The manifold-level statement requires Mathlib's spin-bundle infrastructure which is not yet available; the structure formalizes the algebraic content without re-deriving the PDE asymptotics.

A Python mirror in `src/heat_kernel/` provides numerical evaluators for a_0, a_2, a_4 via `seeley_dewitt_coefficients`, the calibration check via `a2_calibration_passes`, and a 36-case `pytest` suite cross-checking each Lean theorem against the Christensen–Duff rational table. Canonical formulas live in `src/core/formulas.py` as `seeley_dewitt_a0`, `seeley_dewitt_a2_R_coefficient`, `seeley_dewitt_a4_basis`, `G_N_from_seeley_dewitt`, and `heat_kernel_a2_matches_GN_sakharov`, each with explicit `Lean:` cross-references to the corresponding theorem.

VIII. CONCLUSION

The Sakharov–Adler induced-gravity coefficient $G_N^{\mathrm{Sakharov}} = 12\pi/(N_f \Lambda_{\mathrm{UV}}^2)$ is recovered *exactly* from the heat-kernel a_2 coefficient of the free Dirac operator. The formalized result is sharper than a numerical match: it is a biconditional in which any deviation from the textbook Christensen–Duff rational $-1/12$ is detectable via the ADW dimensionless coefficient $\alpha_{\mathrm{ADW}} \neq 1$ in the linearized 6a.1 calculation, and the mismatch factors entirely through the calibration relation rather than through the cutoff or species count. This makes the heat-kernel a_2 calibration a falsifiable mean-field consistency anchor for the rest of Phase 6e: subsequent waves (higher-curvature structure, non-linear diff-invariance, full Einstein equations, microscopic-to-macroscopic coefficient match, Einstein–Cartan extension) inherit a load-bearing cross-bridge to the linearized 6a.1 module that this Wave 1 closes with explicit Lean theorems. Whether the bridge survives the Vergeles unitarity computation will fix the mean-field validity boundary in the natural-parameter band; either outcome is substantive.

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